

A Location-Triggered Augmented Reality Walking Tour Using Snap Spectacles 2021

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ABSTRACT

We present an on-site 3D-animated audiovisual tour guide augmented reality application developed for Snap Spectacles 2021. The primary goal of this project is to explore how to use this experimental product to create an augmented reality tour guide. In addition, we present the design considerations for the user interface and the underlying system architecture. We illustrate the workflow of the tour application and discuss our experience working with Spectacles 2021 and its experimental API. We also present our design choices and directions for future work.

Index Terms: Human-centered computing—Human Computer Interaction—(HCI) Interaction paradigms—Mixed / augmented reality; Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality

1 INTRODUCTION

Most augmented reality (AR) and virtual reality (VR) systems are bulky and cumbersome, making them uncomfortable and impractical for prolonged usage, particularly in outdoor environments. Hence, there has been an increasing interest in developing highly portable computer eyewear that make outdoor use more practical. We explore how Snap Spectacles 2021—an experimental non-commercially available product—can be used to prototype an outdoor AR experience: a location-triggered AR walking tour.

A number of researchers have developed experimental on-site AR tour guides. Höllerer et al. [6] used AR to create situated documentaries about historical events on a college campus, while Vlahakis et al. [11] developed Archeoguide, which helps tourists navigate a cultural heritage site in Olympia, Greece. Similar applications have been created for Pompeii [10] and Edinburgh [4].

We developed an experimental on-site AR tour guide for the Columbia University campus using Spectacles 2021. Columbia University currently provides several kinds of guided tours, including guide-directed tours, self-guided walking tours and virtual tours that present historical information about landmarks. Unlike these tours, we used Spectacles 2021 to develop an AR tour guide that integrates an animated 3D virtual character and historical context with a synchronized audio narration.

2 DESIGN CONSIDERATIONS

We have tried to create an intuitive, consistent, and immersive augmented tour guide that allows visitors to navigate the campus easily on their own. During the tour, visitors are encouraged to explore the campus independently. When they are in the proximity of a landmark, they are guided toward it by navigation arrows. These arrows are triggered to spawn within a fixed radius of the landmark location

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Figure 1: AR tour guide explaining significance of landmark using historical video content overlaid on surrounding environment. Captured from the Spectacles 2021 camera feed.

when the user is near it. In addition, we render an interactive 3D character (a virtual tour guide, shown in Figure 1), accompanying video and audio upon reaching the location.

2.1 Interaction

To interact with the application, we considered both touch and hand-tracking. Some studies demonstrated an increase in user immersion and engagement with hand-tracking [7]. In addition, Meier et al. [8] propose a hybrid approach that uses rapid touch interaction on surfaces in conjunction with headset-tracked hand poses to trigger input in VR. While we wanted to create a more immersive experience using a hybrid interaction modality, the Spectacles 2021 hand-tracking system was not sufficiently precise. Therefore, since we wanted the user to access and view suggested landmarks to explore, we chose well-established touch interaction to allow us to create an intuitive and consistent user interface.

2.2 Navigation and Guide

Since Spectacles 2021 use smartphone-based satellite tracking, which is limited in its accuracy, we needed to ensure that our application provided easy navigation and prevented false location triggers. We decided to introduce redundancy by having different navigation cues. For instance, we used brightly colored landmark arrows that spawn at the landmark location, encouraging the user to head in the direction of the landmark. We also use navigation arrows to guide users to the location and orient them correctly. Furthermore, since we used Snap's experimental location features [3], handling errors, such as a landmark not being detected, was critical. To address these issues, we allowed overriding location triggers, so that users could access the AR content of a landmark by tapping the Spectacles 2021 touchpad.

2.3 Improving Performance

We used Snap's Endurance Mode [2] to temporarily turn the display off between two consecutive landmark locations while exploring. This prevents any visual hindrance while navigating through the

environment. We also play soft background music during Endurance Mode to keep the user engaged while the display is off. Endurance Mode allows us to conserve the Spectacles 2021 battery, which is critical to the experience, as the maximum life of Spectacles 2021 when fully charged is roughly 30 minutes.

2.4 Content Selection

Our objective in content selection was to use multidimensional AR components to maximize immersion. We focused on visual appeal, entertainment, and enjoyment, as Han et al. [5] showed that these components affect supportive behavior through AR satisfaction and experiential authenticity. In addition, previous work in social presence in mixed reality showed that realism and agent interactivity improved engagement [9]. Thus, we use an animated 3D tour guide (adapted from an existing application [1]) that mimics human gestures and emotions.

3 SYSTEM OVERVIEW AND DESIGN

Figure 2 shows the architecture of our system.

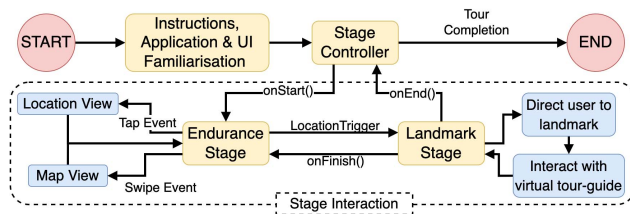


Figure 2: Tour system architecture. The application toggles between Endurance and Landmark stages, facilitating navigation and immersive experience, respectively.

System execution begins with the START state, which initializes the application controller and instantiates the Stage Controller object, loading all the landmark locations. Once the system turns on, the user is given an onboarding tutorial for the system UI. The user is shown a small introductory video of the Morningside Heights campus of Columbia University, encouraging them to explore the campus by themselves.

After the introduction has finished, the system shifts into the Endurance stage, which enables Endurance Mode and temporarily turns off the screen. If the user arrives in the vicinity of a landmark location, the system enters the Landmark stage. The user sees a green landmark arrow (Figure 3a), which suggests they head toward it. Once closer, they see navigation arrows (Figure 3b) that guide them toward the landmark location, where they interact with the virtual tour guide (Figure 3c). The system also uses audio cues and spatial 3D sound, inviting users to follow the arrow and directs their attention toward the tour guide. This sequence is repeated until all landmarks are explored.

4 DISCUSSION

This was a challenging application to implement on a small wearable device. We used experimental APIs, including Location Trigger, which frequently caused spontaneous application shutdowns and imprecise location detection. Hardware limitations included a small (26.3°) diagonal field of view, 30-minute battery life, a small (8 MB) maximum size for the supported application, and heat-induced shutdowns. These constrained our design possibilities and application development process.

5 CONCLUSIONS AND FUTURE WORK

We presented an on-site campus AR tour developed using Snap Spectacles 2021, an experimental, non-commercially available, lightweight, untethered AR eyewear computer. Our future work

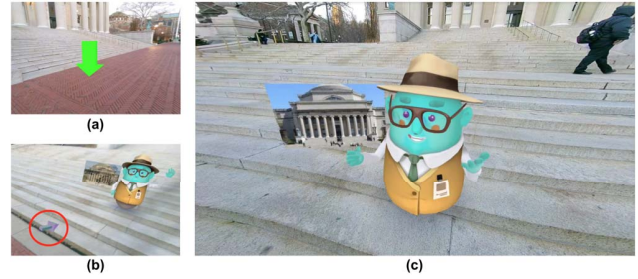


Figure 3: (a) Landmark arrows indicate potential landmark location. (b) Navigation arrow (circled in red) guides user to AR tour guide. (c) Interactive AR tour guide, video, and audio components. Captured from the Spectacles 2021 camera feed.

will explore adding augmentations that are tightly integrated with landmarks and supporting verbal interactions with the tour guide. Our application allows for a high degree of customization and scalability, and we are interested in the possibility of repurposing and scaling it for other outdoor AR experiences.

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